ON FORCE-FREE MAGNETIC FIELDS*

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ABSTRACT

A method of deriving the general solution of the equation curl $\boldsymbol{H} = \alpha\boldsymbol{H}$, where α is constant, is given, the solution being found as the sum of a poloidal and a toroidal vector field. Applying this solution to force-free magnetic fields between spherical boundaries, we find that, as in the axisymmetric case, there is, quite generally, equipartition of energy between poloidal and toroidal components of the field Finally, it is pointed out that any force-free field of this type can decay freely in a fluid without causing any motion of the fluid.

I. INTRODUCTION

The equation curl $\boldsymbol{H} = \alpha\boldsymbol{H}$, characterizing force-free magnetic fields, has recently been discussed by several authors (Lüst and Schlüter 1954; Chandrasekhar 1956*a*, *b*) and the solution in the axisymmetric case for α = Constant has been given (Chandrasekhar 1956*a;* this paper will be referred to as "Paper I"). In this paper we shall give a simpler solution of the equation which will exhibit the mathematical structure of the solution in the axisymmetric case and will include non-axially symmetric solutions as well.

II. THE FUNDAMENTAL EQUATIONS

One possible form of a magnetic field $\boldsymbol{H}$ which will not disturb the equilibrium of a stationary fluid is a solution of the equation

$$\text{curl}\ \boldsymbol{H} = \alpha\boldsymbol{H}\ , \tag{1}$$

where α is a constant. From this equation we can immediately derive the equation

$$\text{curl}^2\ \boldsymbol{H} = \alpha^2\boldsymbol{H}\ , \tag{2}$$

or, since div $\boldsymbol{H} = 0$,

$$\nabla^2\boldsymbol{H} + \alpha^2\boldsymbol{H} = 0\ , \tag{3}$$

which is the vector-wave equation. This step was taken by Namikawa (1955) to solve an equation similar to equation (1) obtained after a separation of the variables in cylindrical polar co-ordinates. Every solution of equation (3) is not necessarily a solution of equation (1), although the converse is true. This indicates that the solution of equation (1) is to be found among the solutions of equation (3).

If ψ is a scalar function such that

$$\nabla^2\psi + \alpha^2\psi = 0\ , \tag{4}$$

then it is well known that three independent solutions of equation (3) are given by

$$\boldsymbol{L} = \text{grad}\ \psi\ , \qquad \boldsymbol{T} = \text{curl}\ (\boldsymbol{a}\psi)\ , \qquad \text{and} \qquad \boldsymbol{S} = \frac{1}{\alpha}\ \text{curl}\ \boldsymbol{T}\ , \tag{5}$$

* The results in this paper were derived independently by the two authors; and they agreed to write it as one.—V. C. A. FERRARO.

where $\boldsymbol{a}$ is a fixed unit vector. It can then be shown that

$$\operatorname{curl} \boldsymbol{S} = aT \,. \tag{6}$$

Equations (5) and (6) then give

$$\operatorname{curl} (\boldsymbol{S}+\boldsymbol{T}) = a (\boldsymbol{S}+\boldsymbol{T}) \,. \tag{7}$$

Therefore, the most general solution of equation (1) is

$$\boldsymbol{H} = \frac{1}{a} \operatorname{curl}^2 (\boldsymbol{a}\psi) + \operatorname{curl} (\boldsymbol{a}\psi) \,. \tag{8}$$

It can easily be verified that $\boldsymbol{H}$ then satisfies equation (1).

The set of fundamental solutions that we shall use are given by replacing $\boldsymbol{a}$ by $\mathbf{1}_r$, the unit vector in the radial direction from the origin, and using the fundamental solutions of equation (4) given by

$$\psi_n^m = Z_n (a r) P_n^m (\cos \theta) \begin{matrix} \cos m\phi \\ \sin m\phi \end{matrix} \,, \tag{9}$$

where

$$Z_n (a r) = \left(\frac{\pi}{2 a r}\right)^{1/2} \mathfrak{C}_{n+1/2} (a r) \,, \tag{10}$$

where $\mathfrak{C}_{n+1/2}(ar)$ is a linear combination of the Bessel functions, $J_{n+1/2}$ and $J_{-(n+1/2)}$, and P_n^m is the Legendre function of order m and degree n. Using spherical polar co-ordinates (r, θ, ϕ), we obtain

$$\boldsymbol{H} = \sum_{n,\,0}^{\infty} \sum_{m=0}^{\infty} a_{m,\,n} \boldsymbol{H}_n^m \,, \tag{11}$$

where

$$\boldsymbol{H}_n^m = \boldsymbol{S}_n^m + \boldsymbol{T}_n^m \tag{12}$$

$$\begin{aligned} \boldsymbol{S}_n^m = & \left[\frac{n(n+1)}{a r} Z_n (a r) P_n^m (\cos \theta) \begin{matrix} \cos m\phi \\ \sin m\phi \end{matrix}\right] \mathbf{1}_r \\ & + \left\{\frac{1}{a r} \frac{d}{d r} [r^2 Z_n (a r)] \frac{\partial}{\partial \theta} P_n^m (\cos \theta) \begin{matrix} \cos m\phi \\ \sin m\phi \end{matrix}\right\} \mathbf{1}_\theta \\ & \mp \left\{\frac{m}{a r \sin \theta} \frac{d}{d r} [r Z_n (a r)] P_n^m (\cos \theta) \begin{matrix} \sin m\phi \\ \cos m\phi \end{matrix}\right\} \mathbf{1}_\phi \,, \end{aligned} \tag{13}$$

and

$$\begin{aligned} \boldsymbol{T}_n^m = & \left[\mp \frac{m}{\sin \theta} Z_n (a r) P_n^m (\cos \theta) \begin{matrix} \sin m\phi \\ \cos m\phi \end{matrix}\right] \mathbf{1}_\theta \\ & - \left[Z_n (a r) \frac{\partial}{\partial \theta} P_n^m (\cos \theta) \begin{matrix} \cos m\phi \\ \sin m\phi \end{matrix}\right] \mathbf{1}_\phi \,. \end{aligned} \tag{14}$$

The functions $\boldsymbol{S}_n^m$ and $\boldsymbol{T}_n^m$ fulfil the usual orthogonality relationships, and

$$\int_0^\pi \int_0^{2\pi} |\boldsymbol{T}_n^m|^2 \sin \theta d\theta d\phi = \frac{2\pi}{2n+1} \frac{(n+m)!}{(n-m)!} n(n+1) Z_n^2 (a r) \,. \tag{15}$$

Since $C_{k+1}^{3/2}(\mu) = dP_n(\mu)/d\mu$, where $C_{k+1}^{3/2}$ is Gegenbauer's function of order $\frac{3}{2}$ and degree $(k+1)$, the results of Paper I are given by equation (11) in the case $m = 0$.

These results show that, given any poloidal field $\boldsymbol{S}$, we can find a toroidal field, $\boldsymbol{T}$, which, when added to $\boldsymbol{S}$ makes the total field $\boldsymbol{S}+\boldsymbol{T}$ force-free. The converse is also true.

III. THE BOUNDARY CONDITIONS

There are no alterations in the boundary conditions discussed in Paper I. Briefly these are as follows. Since, at a discontinuity in α, both $\boldsymbol{H}$ and curl $\boldsymbol{H}/4\pi$ (the current density) are continuous, the normal component of $\boldsymbol{H}$ must vanish at the boundary. The other boundary conditions are that the tangential resolutes of $\boldsymbol{H}$ are continuous at the boundary. An interesting case is when one boundary is a sphere $r = R$. Equations (12), (13), and (14) then show that the boundary conditions are

$$Z_n(\alpha R) = 0 \tag{16}$$

and

$$\frac{1}{\alpha r}\frac{d}{dr}[rZ_n(\alpha r)] \text{ is continuous at the boundary.} \tag{17}$$

Therefore, on any spherical boundary,

$$\boldsymbol{T}_n^m = 0 \text{ (on surface)} . \tag{18}$$

IV. THE ENERGY OF THE MAGNETIC FIELD

We shall now prove that there is equipartition of energy between the toroidal field and the poloidal field in any system with spherical boundaries only. Because $\boldsymbol{S}_n^m$ is orthogonal to $\boldsymbol{T}_n^m$ when integrated over the unit sphere, the total energy in the magnetic field is the sum of the energies contained in the poloidal and toroidal fields. The energy in the poloidal field is

$$E_S = \frac{1}{8\pi}\iiint |\boldsymbol{S}_n^m|^2 d\tau , \tag{19}$$

where the integration is effected over the relevant volume. Using equation (5), we have

$$\begin{aligned} E_S &= \frac{1}{8\pi}\iiint \alpha^{-2} \text{ curl } \boldsymbol{T}_n^m \cdot \text{curl } \boldsymbol{T}_n^m d\tau \\ &= \frac{\alpha^{-2}}{8\pi}\iint_{\text{surface}} \boldsymbol{T}_n^m \times \text{curl } \boldsymbol{T}_n^m d\tau + \frac{\alpha^{-2}}{8\pi}\iiint \boldsymbol{T}_n^m \cdot \text{curl}^2\, \boldsymbol{T}_n^m d\tau . \end{aligned}$$

Finally, using equations (18) and (2), and the fact that the only boundaries are spheres, we obtain

$$E_S = \frac{1}{8\pi}\iiint |\boldsymbol{T}_n^m|^2 d\tau = E_T , \tag{20}$$

the energy in the toroidal field. The total energy, on using equation (15), becomes

$$H_0^2 \frac{n(n+1)}{4n+2}\frac{(n+m)!}{(n-m)!}\int Z_n^2(\alpha r)\, r^2 dr . \tag{21}$$

The value of the energy when $m = 0$ has been given in Paper I for the case when the field lies between two concentric spheres.

V. FREE DECAY

The force-free field is the only stable form of magnetic field which can decay without causing material motions in a fluid. The usual modes of decay associated with a solid conducting sphere would cause motions in a fluid that would probably enhance the rate of decay.

The equations of free decay without material motion, with the conductivity σ measured in e.m. units, are

$$\frac{\partial \boldsymbol{H}}{\partial t} = -\frac{1}{4\pi\sigma} \nabla^2 \boldsymbol{H}, \tag{22}$$

which, on using equation (1), becomes

$$\frac{\partial \boldsymbol{H}}{\partial t} = -\frac{\alpha^2}{4\pi\sigma} \boldsymbol{H}, \tag{23}$$

the solution of which is

$$\boldsymbol{H} = \boldsymbol{H}_0 e^{-(\alpha^2/4\pi\sigma)t}, \tag{24}$$

where $\boldsymbol{H}_0$ is any solution of equation (1) which is not time-dependent. In the case of a sphere, αR is any zero of $J_{n+1/2}(x)$, so that this special case is in agreement with the more general case. It follows that a force-free field, if left to decay freely, will not change its lines of force or current streamlines and so will cause no motion of the field.

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